

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	C0-4	Assessment:	Assessment Tool 1– scenario based problems
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	SLOT-D	Date:	20-08-2026

Code of Conduct

I SYED MUHAFIZ A(192472282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1. Higher Education — Student Satisfaction with Online Classes

A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Given Information

- University's claim: $p \geq 80\%$ (at least 80% of students are satisfied)
- 95% Confidence Interval from the survey: (74%, 79%)
- Claimed / hypothesized value, $p_0 = 80\%$

Step 1: State the Hypotheses

Because the university's claim is a lower-bound ("at least") statement, the hypotheses are set up as a one-sided test:

- H_0 (Null Hypothesis): $p \geq 0.80$ — the university's claim is true
- H_1 (Alternative Hypothesis): $p < 0.80$ — the true satisfaction proportion is lower than claimed

Step 2: Decision Rule

A confidence interval can be used to evaluate a claim about a population proportion. If the claimed value p_0 lies inside the confidence interval, the claim is considered plausible and H_0 is not rejected. If p_0 lies completely outside the interval — and, in particular, if the entire interval lies on one side of p_0 — this is strong evidence against the claim in that direction.

Step 3: Compare the Claimed Value to the Interval

Quantity	Value
Claimed proportion, p_0	80%
Lower limit of 95% CI	74%
Upper limit of 95% CI	79%

Is $p_0 = 80\%$ inside (74%, 79%)?	No — 80% is above the upper limit
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Since the upper limit of the confidence interval (79%) is less than the claimed value of 80%, the entire plausible range for the true satisfaction proportion lies below the university's claimed figure.

Step 4: Conclusion

Result: Because the claimed value of 80% falls outside the 95% confidence interval (74%, 79%), and the entire interval lies below 80%, we reject H_0 . There is statistically significant evidence at the 5% significance level that fewer than 80% of students are actually satisfied with online classes.

Interpretation

The survey data does not support the university's claim. Based on the sample, the true proportion of satisfied students most plausibly lies somewhere between 74% and 79% — a gap of 1 to 6 percentage points below the advertised figure of 80%. While this may seem like a small numerical difference, because the entire confidence interval falls under the threshold, the shortfall is unlikely to be due to random sampling variation alone. The university should treat this as a genuine signal that satisfaction with online classes is lower than reported, investigate possible causes (course design, technical issues, lack of interaction, etc.), and avoid overstating satisfaction levels in official communications.

Statistical Note — One-Sided Claims and Confidence Intervals

Two-sided confidence intervals are commonly used as an approximate check even for one-sided ("at least" / "at most") claims: if the claimed boundary value falls entirely outside the interval on the unfavorable side, this provides strong evidence against the claim. For a fully rigorous one-sided test, a one-sided confidence bound (rather than a two-sided interval) would ideally be constructed, but the conclusion here is unambiguous because the entire two-sided interval already lies below 80%.

Question 2. Mobile Applications — Average Daily Usage Time

A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Given Information

- Developer's claim: $\mu = 30$ minutes per day
- 99% Confidence Interval from the sample: (31.5 minutes, 35.2 minutes)
- Claimed / hypothesized value, $\mu_0 = 30$ minutes

Step 1: State the Hypotheses

- H_0 (Null Hypothesis): $\mu = 30$ minutes — the developer's claim is correct
- H_1 (Alternative Hypothesis): $\mu \neq 30$ minutes — the true average usage time differs from the claim

Step 2: Decision Rule

If the claimed mean μ_0 lies within the confidence interval, the claim is considered a plausible value of the population mean and H_0 is not rejected. If μ_0 lies outside the interval, the claim is not supported by the data and H_0 is rejected.

Step 3: Compare the Claimed Value to the Interval

Quantity	Value
Claimed mean, μ_0	30 minutes
Lower limit of 99% CI	31.5 minutes
Upper limit of 99% CI	35.2 minutes
Is $\mu_0 = 30$ inside (31.5, 35.2)?	No — 30 is below the lower limit

The claimed value of 30 minutes lies below the lower limit of the interval (31.5 minutes), so it does not fall within the 99% confidence interval.

Step 4: Conclusion

Result: Since 30 minutes lies outside the 99% confidence interval (31.5, 35.2), we reject H_0 . The data does NOT support the developer's claim — actual average daily usage time is significantly higher than 30 minutes.

Interpretation

The sample evidence indicates that users actually spend, on average, somewhere between 31.5 and 35.2 minutes per day on the application — noticeably more than the claimed 30 minutes. Because this conclusion is based on a 99% confidence interval (a stricter standard than the more common 95% level), the evidence against the original claim is quite strong. From a business standpoint this is a favorable discrepancy: engagement appears higher than advertised. The developer may wish to update marketing and reporting figures to reflect the more accurate (and more impressive) usage statistic, while also investigating whether this upward shift is consistent across user segments and time periods.

Statistical Note — Effect of Confidence Level on Interval Width

Raising the confidence level from 95% to 99% widens the interval (because a larger critical z-value is used), which normally makes it harder for a hypothesis test based on the interval to reject H_0 . Here, however, the claimed value of 30 minutes is far enough outside the interval that the conclusion still holds decisively even at the stricter 99% level.

Question 3. Financial Services — Fraud Detection System Accuracy

A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Given Information

- Company's claim: the system detects $p = 95\%$ of fraudulent transactions
- Calculated p-value from the hypothesis test = 0.02
- Significance level, $\alpha = 0.05$ (5%)

Step 1: State the Hypotheses

- H_0 (Null Hypothesis): $p = 0.95$ — the detection rate matches the claimed 95%
- H_1 (Alternative Hypothesis): $p \neq 0.95$ — the true detection rate differs from the claimed 95%

Step 2: Understand the p-value

The p-value represents the probability of observing a test result at least as extreme as the one obtained from the sample, assuming that the null hypothesis ($H_0: p = 0.95$) is actually true. A small p-value means the observed data would be quite unlikely if the company's claim were correct.

Step 3: Decision Rule

Reject H_0 if $p\text{-value} \leq \alpha$; Fail to reject H_0 if $p\text{-value} > \alpha$

Step 4: Apply the Decision Rule

Quantity	Value
p-value	0.02
Significance level, α	0.05
Is $p\text{-value} \leq \alpha$?	Yes ($0.02 \leq 0.05$)

Decision	Reject H_0
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Step 5: Conclusion

Result: Since the p-value (0.02) is less than the significance level (0.05), we reject H_0 . There is statistically significant evidence at the 5% level that the fraud detection system's true detection rate differs from the claimed 95%.

Interpretation

The data scientist should conclude that the sample data is inconsistent with the company's claim of a 95% fraud detection rate — the test result would be unlikely to occur (only a 2% chance) if the true detection rate really were 95%. This does not by itself reveal whether the actual performance is better or worse than claimed; that would require examining the sample proportion itself or constructing a confidence interval around it. Given the high stakes of fraud detection (missed fraud has direct financial cost, while false positives inconvenience customers), the company should investigate the discrepancy further, audit the system's real-world performance, and avoid making marketing or regulatory claims that are not statistically supported.

Statistical Note — Type I Error and Significance Level

Choosing $\alpha = 0.05$ means the data scientist accepts a 5% risk of rejecting a true null hypothesis (a Type I error, i.e. wrongly concluding the detection rate differs from 95% when it actually is 95%). Because the p-value (0.02) is comfortably below this threshold, the risk of this particular rejection being a false alarm is relatively low, which strengthens confidence in the conclusion.

Question 4. E-commerce — New Recommendation Algorithm

An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given Information

- Baseline (existing algorithm) conversion rate = 12%
- Observed conversion rate with new algorithm = 15%
- Calculated p-value = 0.03
- Significance level, $\alpha = 0.05$ (5%)

Step 1: State the Hypotheses

Since the company is specifically interested in whether the new algorithm improves (increases) the conversion rate, this is a one-sided (one-tailed) test:

- H_0 (Null Hypothesis): $p_{\text{new}} \leq 0.12$ — the new algorithm does not improve conversion
- H_1 (Alternative Hypothesis): $p_{\text{new}} > 0.12$ — the new algorithm significantly improves conversion

Step 2: Decision Rule

Reject H_0 if $p\text{-value} \leq \alpha$; Fail to reject H_0 if $p\text{-value} > \alpha$

Step 3: Apply the Decision Rule

Quantity	Value
p-value	0.03
Significance level, α	0.05
Is $p\text{-value} \leq \alpha$?	Yes ($0.03 \leq 0.05$)

Decision	Reject H_0
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Step 4: Conclusion

Result: Since the p-value (0.03) is less than α (0.05), we reject H_0 . The increase in conversion rate from 12% to 15% is statistically significant — the new recommendation algorithm significantly improves conversion at the 5% significance level.

Interpretation

The observed 3-percentage-point increase (12% $\rightarrow$ 15%) is unlikely to have occurred purely by chance if the new algorithm truly had no effect, since such a result (or one more extreme) would happen only about 3% of the time under the null hypothesis. In relative terms, this represents a 25% increase in conversion rate ($15/12 = 1.25$), which could translate into a meaningful revenue gain depending on the platform's traffic and order value. Before a company-wide rollout, the team should also consider practical significance alongside statistical significance — factors such as the cost of maintaining the new algorithm, its performance across different customer segments, and whether the improvement is sustained over time — but from a purely statistical standpoint, the evidence supports adopting the new algorithm.

Statistical Note — One-Tailed vs. Two-Tailed Tests

Because the business question here is specifically about improvement (an increase), a one-tailed test is appropriate and uses the full significance level α in one direction. This differs from Questions 6–8, where two-sided tests were used because deviation in either direction (higher or lower than claimed) was of interest. Using the correct tail for the hypothesis affects how the p-value is calculated and interpreted, so it is important to match the test direction to the actual business question.

Question 5. Healthcare — AI-Based Patient Scheduling System

A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Given Information

- Goal: determine whether the AI system reduces average patient waiting time
- Calculated p-value = 0.08
- Significance level, $\alpha = 0.05$ (5%)

Step 1: State the Hypotheses

- H_0 (Null Hypothesis): the AI system does not reduce average waiting time ($\mu_{\text{new}} \geq \mu_{\text{old}}$)
- H_1 (Alternative Hypothesis): the AI system reduces average waiting time ($\mu_{\text{new}} < \mu_{\text{old}}$)

Step 2: Decision Rule

Reject H_0 if $p\text{-value} \leq \alpha$; Fail to reject H_0 if $p\text{-value} > \alpha$

Step 3: Apply the Decision Rule

Quantity	Value
p-value	0.08
Significance level, α	0.05
Is $p\text{-value} \leq \alpha$?	No ($0.08 > 0.05$)
Decision	Fail to reject H_0

Step 4: Conclusion

Result: Since the p-value (0.08) is greater than α (0.05), we fail to reject H_0 . There is NOT enough statistical evidence at the 5% significance level to conclude that the AI-based scheduling system significantly reduces average patient waiting time.

Interpretation

Although the sample data shows a trend in the desired direction, the result is not strong enough to rule out chance as an explanation at the conventional 5% significance level. It is worth noting that a p-value of 0.08 would be considered statistically significant at a 10% significance level, so this is a borderline result rather than clear evidence of no effect. The hospital should be cautious about concluding either that the system works or that it does not — instead, this result suggests the possibility of a real but modest improvement that the current sample was not large enough to detect with confidence. Reasonable next steps include collecting more data to increase statistical power, extending the trial period, or examining whether the effect is stronger for specific patient groups or times of day before making a final decision on the system's effectiveness.

Statistical Note — Failing to Reject H_0 Is Not the Same as Proving H_0

"Fail to reject H_0 " means the evidence is insufficient to confidently conclude an effect exists — it does not prove that the AI system has no effect on waiting times. This distinction matters here because of the risk of a Type II error (failing to detect a real improvement due to limited sample size or natural variability). Increasing the sample size is often the most direct way to increase the test's statistical power and reduce this risk in a follow-up study.

Summary of Results

Q.	Scenario	Method	CI / p-value	Conclusion
1	Student Satisfaction	95% CI	(74%, 79%)	80% not plausible — reject H_0
2	App Daily Usage	99% CI	(31.5, 35.2) min	30 min not plausible — reject H_0
3	Fraud Detection Rate	p-value	$p = 0.02 < \alpha = 0.05$	Reject H_0 — differs from 95%
4	Recommendation Algorithm	p-value	$p = 0.03 < \alpha = 0.05$	Reject H_0 — significant improvement
5	AI Scheduling System	p-value	$p = 0.08 > \alpha = 0.05$	Fail to reject H_0 — not significant